Patch-Based Diffusion Models for Solving Inverse Problems in Computed Tomography Imaging

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Abstract

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1 Introduction

Computed Tomography (CT) imaging is widely used in both medicine and industry for investigating the internal structure of objects in a non-destructive manner. It works by directing X-rays through a sample from a number of different angles, and measuring their attenuation at each angle. In the medical domain, we wish to minimise the radiation dose to the patient, so measurements are conducted at as few angles (views) as possible and with the lowest possible X-ray energy. This leads to artifacts from both Gaussian and Poisson noise, as well as from the limited number of views restricting the attainable information, making it difficult to accurately reconstruct the internal structure of the sample. Although industrial applications are often able to use higher energy X-rays and use many more views, there are still many cases where radiation dosage can damage the sample, or where only a limited number of views are possible due to movement [].

In this work, we aim to reproduce work by Ho et. al. Ho et al. [1], which introduces a method for improving the quality of very low-dose CT reconstructions by using a patch-based diffusion (PaDIS) model as a prior. The reproduction of CT and machine learning research is often difficult due to the relatively limited number of public CT datasets, concerns regarding patient privacy, and poorly engineered or maintained codebases. For this reason, we developed our reproduction natively within the LION framework [**?**], which provides physically accurate CT projection operators, standardised datasets and metrics, and a well-engineered codebase for developing and testing machine learning methods for CT reconstruction. We have therefore oped to maximise the utility of this work and enable easy comparison to future work by using a default LION dataset and geometry for our reproduction.

As well as implementing the core PaDIS pipeline, we also re-implemented the patch-averaging [2] and patch-stitching [3] inspired variants discussed in the PaDIS paper. Furthermore, we also ran experiments with some of the methods already implemented in LION, choosing those which are similar to the comparison models used by the original PaDIS paper.

The full methodology of our reproduction is described in **??**, with the results presented in **??**. These are then discussed in **??**, along with comments on the reproducibility of the Hu et. al. paper, before conclusions are presented in **??**. A theoretical overview of CT reconstruction and the variance exploding noise conditional score model (NCSN) used by PaDIS now follows.

2 Background

Here, we present a summary of the theory and literature required to understand the experiments presented in this work. We focus on CT reconstruction, diffusion models and their use in solving inverse problems, and the PaDIS method. We then summarise the literature relevant to PaDIS, including both its motivators and derivatives. The formulation presented in Section 2.1 summarises the key points presented by Kak and Stanley [**?**], and utilises their notation.

2.1 Basics of Computed Tomography

In X-ray CT, an object is illuminated by X-rays from multiple angles, with the intensity of radiation transmitted through the object measured by detectors on the opposite side of the object. Through this method, we aim to reconstruct a spatial distribution of the linear attenuation coefficient, written $f(x,y)$ in the two-dimensional case, and with units of inverse length. This gives the probability per unit path length that an X-ray photon is removed from the beam by absorption or scattering. Therefore, assuming X-rays are monoenergetic, the transmitted intensity along a ray L should obey the Beer-Lambert law,

$$I = I_0 \exp\left(-\int_L f(x,y)\,ds\right), \tag{2.1}$$

where I_0 is the intensity of radiation incident on the object, I is the detected intensity, and ds denotes integration along the ray. Taking logarithms then gives the linearised projection

$$P = -\log\left(\frac{I}{I_0}\right) = \int_L f(x,y)\,ds, \tag{2.2}$$

meaning log-normalised CT measurements are approximated by a line integral through the attenuation image. This approach assumes that beams are narrow, monoenergetic and non-diffracting, and neglects scattering, beam hardening and finite aperture effects, all of which contribute in clinical CT. However, it provides a tractable mathematical basis within which to consider reconstruction algorithms.

2.2 Detector Geometries

2.3 Line Integrals, Projections and Sinograms

For a fixed projection angle θ, we can define a line in the image plane by

$$x\cos\theta + y\sin\theta = t, \tag{2.3}$$

where t is the perpendicular distance of the line from the origin, as shown in Figure **??**. The integral over this line can then be denoted as

$$P_\theta(t) = \int_{L_{(\theta,t)}} f(x,y)\,ds, \tag{2.4}$$

or equivalently in Dirac delta form,

$$P_\theta(t) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\delta(x\cos\theta + y\sin\theta - t)\,dx\,dy. \tag{2.5}$$

Notably, the projection $P_\theta(t)$ is therefore also the Radon transform [**?**] of $f(x,y)$ at angle θ. We collect a line or grid of these projections simultaneously, each corresponding to the same θ alignment, but different values of t. In a hypothetical parallel-beam scan, all of these θ-aligned rays are parallel, whereas in a realistic fan-beam or cone-beam scan, the angle refers to the central projection angle, from which other rays diverge.

In practice, we sample these projection lines from K different angular views, $\theta_1, \theta_2, \ldots, \theta_K$, and collect the projections within N detector bins corresponding to $t_1, t_2, \ldots, t_N$, forming an array of measurements called a sinogram, denoted

$$Y_{j,i} = P_{\theta_i}(t_j), \quad i = 1, \ldots, K, \quad j = 1, \ldots, N. \tag{2.6}$$

Here, each column is a line of projections at directed angle θ, and each row corresponds to one detector bin. By vectorising the image of $f(x,y)$ as $\mathbf{x} \in \mathbb{R}^n$, and the sinogram as $\mathbf{y} \in \mathbb{R}^m$, the discretised CT forward model can be written as

$$\mathbf{y} = A\mathbf{x} + \eta, \tag{2.7}$$

where $A \in \mathbb{R}^{m\times n}$ is the discretised forward projection operator, η encodes noise and modelling errors, and $m = KN$ in 2D. We can then express this operator A explicitly by denoting an image pixel ℓ by Ω_ℓ, and using the pixel basis function

$$\phi_\ell(x,y) = \begin{cases} 1, & (x,y) \in \Omega_\ell, \\ 0, & \text{otherwise}, \end{cases} \tag{2.8}$$

such that we can approximate the attenuation field as

$$f(x,y) \approx \sum_{\ell=1}^{n} x_\ell \phi_\ell(x,y), \tag{2.9}$$

where x_ℓ is the value of pixel ℓ. Therefore, substituting into the line integral and comparing with Equation (2.7) gives

$$P_{\theta_i}(t_j) \approx \sum_{\ell=1}^{n} x_\ell \int_{L_{(\theta_i,t_j)}} \phi_\ell(x,y)\, ds, \tag{2.10}$$

$$\implies A_{(j,i),\ell} = \int_{L_{(\theta_i,t_j)}} \phi_\ell(x,y)\, ds, \tag{2.11}$$

so each row of A corresponds to a single measured ray, with entries encoding the path length of that ray within each image pixel. This matrix is generally extremely large, and impractical to store or manipulate.

2.4 The Fourier Slice Theorem

Using the perpendicular distance of a ray from the origin, Equation (**??**), and the distance along that ray given by

$$s = -x\sin\theta + y\cos\theta, \tag{2.12}$$

give the projection at angle θ as

$$P_\theta(t) = \int_{-\infty}^{\infty} f(t\cos\theta - s\sin\theta,\, t\sin\theta + s\cos\theta)\, ds. \tag{2.13}$$

Taking the Fourier transform of this with respect to t then gives

$$S_\theta(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t\cos\theta - s\sin\theta,\, t\sin\theta + s\cos\theta) e^{-2\pi i\omega t}\, ds\, dt, \tag{2.14}$$

so converting to the (x,y) coordinate system gives

$$S_\theta(\omega) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)e^{-2\pi i\omega(x\cos\theta + y\sin\theta)}\,dx\,dy = F(\omega\cos\theta, \omega\sin\theta). \tag{2.15}$$

Therefore, the Fourier transform of a projection gives the two-dimensional Fourier transform of the object along the radial line at the same angle.

2.5 Dosage and Noise

The radiation dose imparted on a patient is directly proportional to the total energy deposited in the tissue. For this reason, it is also directly proportional to the energy of the photons used, the number of photons released at each view, and the number of views acquired. Dosage can therefore be reduced by decreasing any of these properties. However, these each decrease imaging quality.

Measurement noise is intrinsically linked to the intensity of radiation used. If N_0 photons are incident on the sample along a ray with projection value p, then we expect to detect

$$\bar{N} = N_0 \exp(-p). \tag{2.16}$$

As this is a counting experiment, the noise due to counting errors can be approximated as Poisson such that

$$N \sim \text{Poisson}(\bar{N}). \tag{2.17}$$

The projection estimate is

$$\hat{p} = -\log\left(\frac{N}{N_0}\right), \tag{2.18}$$

so at high photon counts where $\text{Poisson}(\bar{N}) \approx \mathcal{N}(\bar{N}, \bar{N})$, Taylor expansion gives

$$\hat{p} \approx p + \eta, \quad \eta \sim \mathcal{N}\left(0, \frac{1}{\bar{N}}\right). \tag{2.19}$$

At low photon counts, however, we may have $N = 0$, in which case $\hat{p} = -\log 0$, which is undefined. Furthermore, relative fluctuations are large, preventing Taylor expansion, and in general, the noise after the logarithm becomes non-Gaussian, signal-dependent, biased and possibly undefined [], meaning this domain is often better to model with the Poisson likelihood. Decreasing intensity therefore increases the difficulty in modelling, and is not discussed in detail in this work. However, interesting advances in using diffusion models trained with these sorts of priors can be found at [].

Decreasing the number of views also decreases the dosage, but in accordance with the Fourier slice theorem, this also limits...

3 Methods

4 Results

5 Discussion

6 Conclusion

References

[1] Ho, J., Jain, A., and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models.

[2] Özdenizci, O. and Legenstein, R. (2023). Restoring Vision in Adverse Weather Conditions With Patch-Based Denoising Diffusion Models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(8):10346–10357.

[3] Rumberger, J. L., Yu, X., Hirsch, P., Dohmen, M., Guarino, V. E., Mokarian, A., Mais, L., Funke, J., and Kainmueller, D. (2021). How Shift Equivariance Impacts Metric Learning for Instance Segmentation. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 7108–7116, Montreal, QC, Canada. IEEE.

A LLM Usage Declaration

LLMs were used for assistance with coding, as declared in the repository README file. They were never used to generate text for inclusion in this document, and were only used to proofread and suggest minor wording alterations.

B Additional Figures